

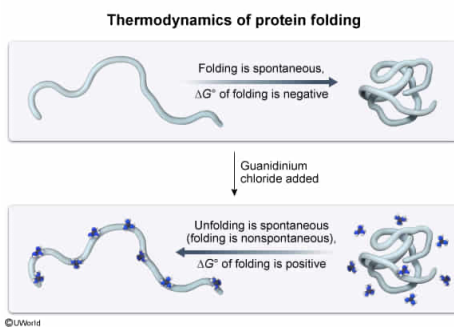
A protein folds spontaneously in phosphate-buffered saline (PBS). Guanidinium chloride is gradually titrated into the PBS-buffered protein solution, causing the protein to denature. During the titration, the ΔG° of protein folding is:

- ✓ ☒ A. initially negative and becomes positive. [68%]
☐ B. initially positive and becomes negative. [14%]
☐ C. initially negative and becomes more negative. [12%]
☐ D. initially positive and becomes more positive. [4%]

Correct

68% answered correctly

Explanation:



The **standard Gibbs free energy (ΔG°)** describes whether a reaction favors products or reactants. Mathematically, it is expressed as $\Delta G^\circ = -RT \ln(K_{eq})$ where R is the gas constant, T is the temperature in Kelvin, and K_{eq} is the equilibrium ratio of products to reactants. When the reaction **favors products**, K_{eq} is greater than 1 and $\ln(K_{eq})$ is positive. Multiplying this positive number by $-RT$ yields a **negative ΔG°** . Conversely, when the reaction **favors reactants**, K_{eq} is less than 1, making $\ln(K_{eq})$ negative and **ΔG° positive**.

Under conditions that mimic a physiological setting, such as phosphate-buffered saline (PBS) solutions, most proteins **spontaneously fold** into a conformation that minimizes interactions between hydrophobic residues and water (**hydrophobic effect**). Because the equilibrium of the protein in PBS lies toward the folded form (products), ΔG° of folding is negative under these conditions.

Addition of guanidinium chloride causes proteins to **denature**, indicating that it **shifts the equilibrium** to favor the **unfolded state** (reactants). Because the equilibrium lies toward reactants in the presence of guanidinium chloride, the ΔG° of folding becomes positive. Therefore, during the titration, the ΔG° of folding is **initially negative and becomes positive**.

(Choice B) The ΔG° of *unfolding* is initially positive and changes to negative, but the ΔG° of *folding* is the opposite.

(Choices C and D) The ΔG° of folding is initially negative but becomes positive.

Educational objective:

The standard Gibbs free energy (ΔG°) describes whether a reaction favors products or reactants. A negative ΔG° indicates that *products* are favored, whereas a positive ΔG° indicates that *reactants* are favored. The equilibrium of a process such as protein folding may be shifted by denaturing agents such that the reaction shifts from favoring folded proteins (products, negative ΔG°) to favoring unfolded proteins (reactants, positive ΔG°).

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